

Cracking the Snake Code on *The Bad Guys*

Yukinori Inagaki
DreamWorks Animation
Glendale, California, USA
yukinori.inagaki@dreamworks.com

Evan Boucher*
Spire Animation Studios
Burbank, California, USA
evan.m.boucher@gmail.com

Min Hong
DreamWorks Animation
Glendale, California, USA
min.hong@dreamworks.com

Ludovic Bouancheau
DreamWorks Animation
Glendale, California, USA
ludovic.bouancheau@dreamworks.com

Fredrik Nilsson†
Epic Games
Los Angeles, California, USA
frednilsson@gmail.com



Figure 1: Shots from *The Bad Guys*: Mr. Snake gives a toast, and Mr. Snake plays the guitar.

ABSTRACT

A snake is simply a tube and doesn't have shoulders or hips that are traditionally complicated to rig on human characters. However, rigging a snake is complicated enough even just for slithering actions. Mr. Snake in the film *The Bad Guys* uses its tail like an arm and acts like a human, walks like a human, and even plays a guitar! This talk presents the wide variety of rigging techniques that were used to achieve the unique cartoony actions of Mr. Snake.

CCS CONCEPTS

• Computing methodologies → Computer graphics.

KEYWORDS

characters, snake, animation

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1 INTRODUCTION

For Master Viper in the film *Kung Fu Panda*, we developed a snake rig that uses a Bezier curve, which we refer to as the Viper System. Animators can insert, remove and animate the control points of the Bezier curve and slide a snake along the curve. This worked well for slithering motion. However, the animation style in *The Bad Guys* is much more exaggerated like hand-drawn 2D animation, and we had a lot of acting shots of Mr. Snake in the film, which was not easy to animate with the Viper System. Thus, we needed to update our snake rig to make it easier to animate acting shots, and capable of posing cartoon style facial expressions.

2 SLITHERING AND ACTING

To make Mr. Snake capable of a variety of actions, we used a combination of three systems: Path System, IK Boxes, and Viper System as shown in Figure 2. The Path System handles slithering motions. IK Boxes are used for acting shots. A blend of those two controls the position and orientation of each Bezier control point of the Viper system. And, the Viper system is used to lock and unlock part of the body in a certain position on the Bezier curve. Also, it manages activation and deactivation of the control points on the curve.

Slithering motion of a snake is achieved by sliding its body along a wavy path. For the *Kung Fu Panda* series, we provided animation controls for editing control points of a Bezier curve and slide amount on the curve for such movement. To animate a snake in such a way, an animator needs to create the curve first. This makes it hard to edit the timing of the action or shape of the curve. So, instead of sliding Mr. Snake on a pre-placed Bezier curve, our Path System uses a motion trail of a joint placed under the head. This simplified

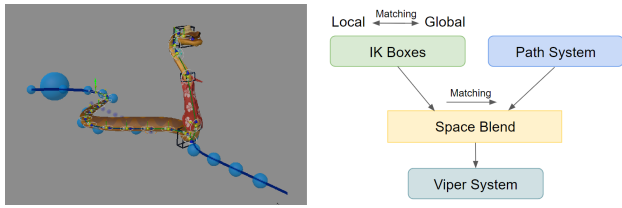


Figure 2: Control icons of the Path System, IK Boxes, and Viper System, and the diagram of how the three systems are layered.

the process of animating slithering motion, and made it easier to increase the appeal of the animation through iteration.

It is important to rig a character to enable animators to reach desired poses with as few animation controls as possible, especially for blocking. For the acting shots of Mr. Snake, we created IK Boxes that have hierarchical structure as shown in Figure 3. IK Boxes consist of five main boxes: Root, TorsoBox, HeadBox, HipBox, and TailBox. Between those boxes, Mid boxes are placed, and get fifty percent influence from the adjacent boxes. Sub boxes are placed between middle boxes and main boxes. And, we placed another layer of two sub boxes around the neck area. Once an animator roughly reaches the desired pose with those box controls, those boxes can be detached from the hierarchy to world space to fine tune the poses without getting influence from other boxes. We provided matching systems for switching those two spaces without changing the snake's pose. Mr. Snake uses its tail as if it is a hand. The TailBox can be used as a wrist control by sliding the tail tip outward, and it can be used as a finger as shown in Figure 1.

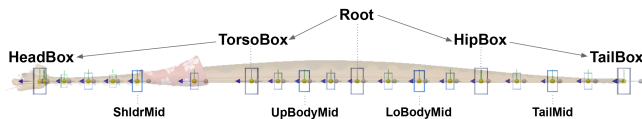


Figure 3: Hierarchical structure of the IK Boxes on Mr. Snake.

An animator can blend any body area between the Path System and IK Boxes as needed. This blend controls the position and orientation of Bezier curve control points in the Viper System. Instead of creating a whole new system, we used the existing Viper System as the final layer of the snake rig to reduce the cost of the development, and utilized the various functionalities in it. For example, the functionality to lock a body part to a certain position on the Bezier curve was a key feature. It was used to lock the base of the neck on the IK box controller there so that the neck can be stretched and squashed. When all the body parts are unlocked, all the joints on the snake body are distributed from the head to tail based on the arc length required to maintain the original length.

3 SHAPES

Facial expressions and silhouette are important elements for conveying the emotion, mood, and traits of a character in the film. For Mr. Snake to achieve the wide range of facial expressions required by the art-direction, we had to adjust our standard face rig. The



Figure 4: Facial expression comparison between before adding specialty face cluster controls on the left and after on the right.

original edge flow from the model didn't have enough resolution in the cheeks, and we were getting a lot of faceted geometry on mouth shapes when the animators pushed it to extremes. By working with the modeling department to up-res the area, and through meticulous sculpting work we were able to get the mouth corner to push all the way to the back of the skull, as well as get it to 'extrude' out of the lower jaw when dropped down, which was a feature requested by the art-direction. This created another challenge. The snake model included the membrane in the mouth to imply the jaw adductor muscles. This created a great bit of character for Mr. Snake, but on certain expressions, it required custom controls to relax the membrane flat into the mouth interior to get it out of the way, so the animators could remove it when needed, but still use it when it was appealing. Mr. Snake had a few bonus controls that the other characters didn't have due to his unique skull shape as well. A number of broadly weighted custom cluster dots were added to allow animators to shape clean and appealing silhouettes for the jaw, cheeks, and eyeballs (since they are on top of the head unlike the other characters). Having a forked tongue, also unique to this character, Mr. Snake required adding a wave system so animators could flick the tongue with a few simple controls.

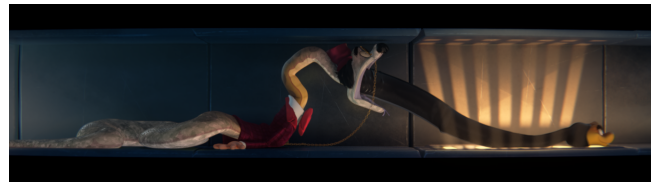


Figure 5: The shot where Mr. Snake exuviates and slides out from the old skin.

4 SECOND SNAKE

The second snake was used in special cases, for example for the shot where Mr. Snake exuviates and slides out of the old skin as shown in Figure 5. The part of the second snake needed to be shrunk down when it was in the original snake's mouth. To accomplish this, we used skin weights that contain parametric positions on the body (tail to be 0 and head to be 1), and used the weight values for hiding from any given threshold towards the tail and/or towards the head. This technique was also used in a shot where Mr. Snake is tied his neck into a knot.